# An Allan Variance Real-Time Processing System for Frequency Stability Measurements of Semiconductor Lasers

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***Abstract*—A real-time frequency stability measurement system for semiconductor lasers was developed. Since the frequency of the input signal is successively measured without clearing the counter, the Allan variance can be accurately measured by following its definition. Measurements of the Allan variance made with this system are more accurate than those made with conventional instruments. The Allan variance may be measured for integration times $\tau$ from 1 μs to 10 000 s, and the number $N$ of measured frequencies averaged over the integration time $\tau$ can be arbitrarily selected up to $N = 707$ for each integration time. The highest measurable frequency was 90 MHz. It was demonstrated experimentally that this system can be used for measurements of the frequency stability of semiconductor lasers.**

## I. Introduction

TEMPORAL coherence of semiconductor lasers needs to be improved when they are used for coherent optical communications, and coherent optical measurements. To accomplish this, several groups have been using frequency stabilization [1], [2] and linewidth reduction of the field spectrum [2]–[4]. They have also been developing a heterodyne frequency-locked loop (HFLL) [5] between semiconductor lasers by using frequency offset locking methods [6]. In order to evaluate the performance of the HFLL, measurements of the frequency stability of the heterodyne signal were required. For longer integration times and an increased number of measurements, frequency stability needs to be efficiently evaluated. A frequency and time interval analyzer HP5371A (Hewlett-Packard) [7] is commercially available for measuring frequency stability. Its fundamental functions are continuous counting and recording of the time of signal zero-crossings. In other words, it measures the relation between time and frequency. Frequency stability measurement does not require the determination of the heterodyne frequency but only the measurement of relative frequency change. The HP5371A can measure the frequency stability for integration times up to 8 s. In the present paper, the authors developed a real-time frequency stability measurement system, i.e., the Allan variance real-time processing system (ARPS) which is quite simple and can measure frequency stability for integration times from 1 $\mu$s to 10 000 s.

Manuscript received February 4, 1989; revised February 26, 1990.

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IEEE Log Number 9036026.

## II. Hardware of the ARPS

The ARPS calculates the square root of the Allan variance as a measure of the frequency stability of semiconductor lasers. The Allan variance is calculated [8] by

$$\sigma_y^2(\tau) = \frac{1}{2(N-1)} \sum_{i=1}^{N-1} (f_{i+1} - f_i)^2 / f_0^2 \qquad (1)$$

where $f_i$ is a signal frequency averaged over the integration time $\tau$, $N$ is the number of averaged frequencies $(f_i)$, and $f_0$ is the nominal frequency of the oscillator that is being evaluated. The averaged frequency $f_i$ can be obtained by

$$f_i = (C_{i+1} - C_i)/\tau \qquad (2)$$

where $C_i$ is the number of TTL pulses obtained from positive-going zero crossings of the sinusoidal analog input signal in the time interval $\tau$ (s). The timing chart is shown in Fig. 1. It should be noted that the definition of the Allan variance requires zero deadtime between successive measurements of $C_i$ [8]. The deadtime of this system is sufficiently small compared to $\tau$. If these measurements were to contain significant deadtime, the Allan variance could diverge as it does when the fluctuations are caused by flicker noise, random walk, etc. [9].

The block diagram of the ARPS is shown in Fig. 2. The ARPS is composed of a counter, a memory, and a personal computer (PC). Since high-speed-type TTL integrated circuits are used in the counter and memory, the maximum measurable frequency of the input signal is as high as 90 MHz. The number of TTL pulses, that is converted from the sinusoidal analog input signal, is continuously counted by a 32-bit synchronous binary counter without clearing the counted value. Thus there is no deadtime in counting the pulse number. The output signal from the counter is transferred to a 32-bit latch. This latch is followed by two 16-bit buffers, and these buffers are connected to the data terminals of a 4-kbyte static RAM. Since this RAM is a 16-bit × 2048 device, output signals from the 32-bit latch are divided into the upper 16-bit and the lower 16-bits. The values of these divided bits are stored in this RAM and the shortest measurable integra-

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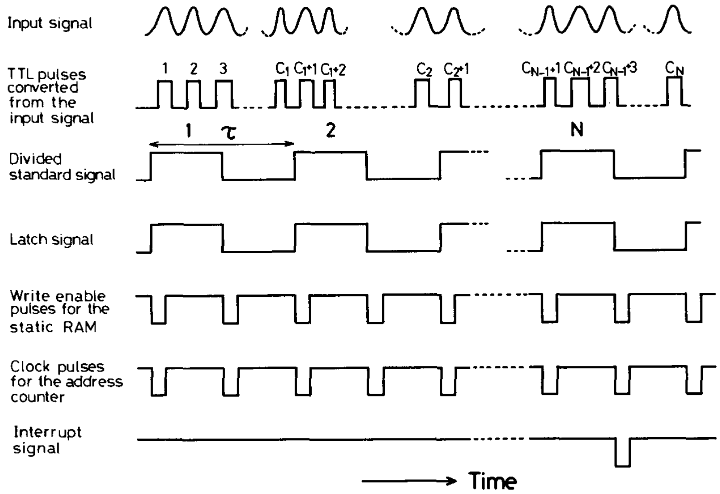


Fig. 1. Timing chart of the TTL pulses used in the ARPS. $C_i$ is a number of the input TTL pulses coming into the counter within the time interval $\tau$, where $\tau$ represents the integration time of the Allan variance measurement. The time interval between adjacent pulses of the divided standard signal determines $\tau$.

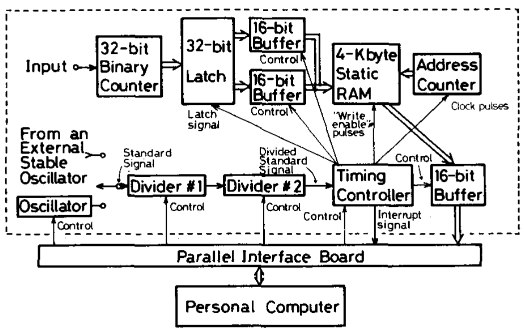


Fig. 2. Block diagram of the ARPS. The block surrounded by the broken lines represents the counter and memory.

tion time of frequency stability is determined by its access speed. Because of the capacity of the RAM, the maximum value of $N$ in (1) can be fixed at 707, which is large enough to measure the Allan variance accurately. An address counter is connected to the address terminals of the static RAM to up-count the stored address. Furthermore, a 16-bit buffer is also connected to the data terminals of the RAM to transfer the counted signal stored in the RAM to the PC.

A standard TTL pulse signal is generated from an oscillator for timing control and for fixing the value of the integration time $\tau$. Its repetition frequency is 1 MHz, and the frequency accuracy and stability are $10^{-4}$ and $10^{-5}$/year, respectively. Two dividers are connected in series to the next stage of the oscillator. The frequency of the standard signal can be divided by anything from 1 to $10^{10}$ by these dividers, where the division ratio is selected by the control signals applied to the dividers from the PC. Thus the frequency is varied between 1 MHz and 0.0001 Hz, and this varies the range of $\tau$ between 1 $\mu$s and 10 000 s. An external standard TTL pulse signal of 1 MHz frequency can also be used instead of the signal from the oscillator. The Allan variance can be measured with higher precision if the external standard signal is derived from a more stable oscillator. If an external standard signal with a frequency lower than 1 MHz is supplied, the frequency stability for $\tau \geqq 10\,000$ s can be measured. In this case, the longest measurable integration time is given by $f_{\text{ext}}^{-1} \times 10^{10}$ (s), where $f_{\text{ext}}$ (Hz) is the frequency of the external standard signal. Control signals in the timing controller for the latch, the static RAM, the address counter, and the buffers can be generated by using the output signal from this oscillator or they can be generated from the oscillator in the PC.

The PC used in the ARPS was a PC-9801Vm (NEC). A parallel interface board with two programmable peripheral interface IC's was added to the PC. This board was used to exchange the signals between the PC and the block of the counter and memory.

## III. Operation of the ARPS

Fig. 1 shows a timing chart of the pulses used in the ARPS. The ARPS begins measuring the values of $f_i$ without any manual operation when the oscillator starts generating the standard signal. The "start/stop" command to the standard signal is controlled by a program of the PC, and the time interval between adjacent pulses of the standard signal is fixed to be the same as the integration time $\tau$ for calculating the Allan variance. The 32-bit counted values of $C_i$ are latched at every $\tau$ (s). The output signals from the latch are divided into two 16 bits, and sent to the two 16-bit buffers. To store these two 16-bit counted values into the static RAM, the timing controller generates two "write enable" pulses and two clock pulses for the static RAM and for the address counter, respectively. This

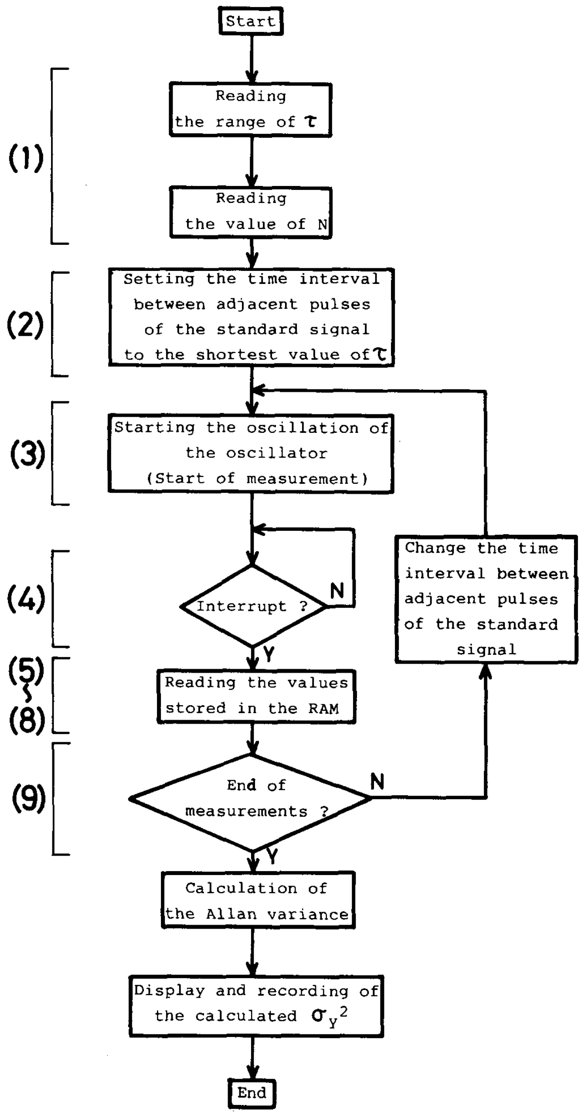


Fig. 3. Flowchart of the program of the ARPS. Each number given next to the block corresponds to sequence number described in the paper.

occurs at each rising edge of the standard signal pulse from the oscillator.

Fig. 3 shows a flowchart of the computer program for the ARPS. Functions of this program are reading the values of the parameters $\tau$ and $N$, initializing the counter and memory, reading the value stored in the RAM, and calculating the Allan variance. The operating procedures of the computer program are summarized as follows.

1) As soon as the input of a range of $\tau$ and $N$ for each $\tau$ is finished, all the flip-flops used in the timing controller and the address counter are cleared.

2) The time interval between adjacent pulses of the standard signal is fixed to the value of the shortest integration time, and an initial value of the address counter is fixed to $(2048 - 2 \times N)$.

3) After initializing by 1) and 2), the oscillator begins generating the standard signal. Data acquisition is then carried out automatically.

4) After the static RAM stores the values of $C_i$ for $N$ times (i.e., when the value of the address counter reaches 2048), an interrupt signal is sent from the block of the counter and memory to the PC.

5) When the interrupt signal is received by the PC, the program stops generating the standard signal from the oscillator, clears all flip-flops, and again fixes the value of the address counter to $(2048 - 2 \times N)$.

6) The program carries out an IN read command through the parallel interface board. The values stored in the memory cells at address numbers $(2048 - 2 \times N)$ and $(2048 - 2 \times N + 1)$ in the static RAM are read by the PC. The TTL level of the input–output read signal (IOR signal) of the PC is changed from high to low to high in sequence by the IN command.

7) By using the IOR signal, the timing controller sends two more clock pulses to the address counter. Then, the values stored in the memory cells at address numbers $(2048 - 2 \times N + 2)$ and $(2048 - 2 \times N + 3)$ in the static RAM are read by the PC by a second execution of the IN command.

8) When the program carries out the IN command $N$ times, all values stored in the RAM are read into the PC.

9) If there are any other integration times $\tau$ for measuring the Allan variance, the time interval between adjacent pulses of the standard signal is fixed to the next $\tau$ and the procedures of 2)–8) are repeated.

The Allan variance is then calculated by using 1) and 2), and the measured values of $C_i$ obtained by the procedures of 2)–9).

## IV. Evaluation of Performances of the ARPS and the Examples of Measurements

Fig. 4 shows a range of values of the square root of the Allan variance which can be measured by the ARPS. On the vertical axis, the values of the squared root of the Allan variance are normalized to the nominal optical frequency $f_0$ (375 T · Hz) of a semiconductor laser of 0.8-$\mu$m wavelength. The range of integration times for which the Allan variance could be measured was from 1 $\mu$s to 10 000 s. Since the value of $C_i$ was measured by the TTL counter in the ARPS, a roundoff error of $\pm 1$ count may occur. Because of this count error, the lower limit of the measurable range of the square root of the Allan variance is expressed as

$$\sigma_y(\tau)_{\text{lower limit}} = f_0^{-1} \cdot \tau^{-1}. \qquad (3)$$

Line $A$ in Fig. 4 represents the value of (3). The upper limit of the measurable range is determined by the response speed (line $B$) of the TTL counter used in the ARPS and by the maximum number of pulses countable (line $C$) by the counter ($2^{32} = 4.3 \times 10^9$ counts). Curve $D$ represents a calculated value of normalized frequency fluctuations caused by the spontaneous emission of 0.8 $\mu$m semiconductor lasers whose typical optical output

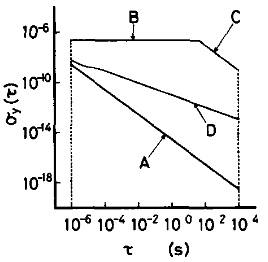


Fig. 4. The range of values of the Allan variance which can be measured by the ARPS. The vertical axis represents the square root of the Allan variance, which is normalized to the optical frequency of a semiconductor laser of 0.8 $\mu$m wavelength. Curve $A$: Lower limit caused by $\pm 1$ count error. Curve $B$: Upper limit determined by the response speed of the TTL counter used in the ARPS. Curve $C$: Upper limit determined by the maximum number of pulses countable by the counter ($2^{32} = 4.3 \times 10^9$ counts). Curve $D$: The calculated value of the frequency stability of 0.8 $\mu$m AlGaAs lasers determined by the magnitude of spontaneous emission [11].

power is 3 mW [10]. Since curve $D$ lies within the range surrounded by the lines $A$, $B$, and $C$, we conclude that the frequency stability of semiconductor lasers can be measured by the ARPS.

To check the operation of the ARPS, a test signal was prepared by dithering the output of a frequency synthesizer to give

$$f(t) = f_0 + (\Delta f/2) \sin (2\pi f_m t) \tag{4}$$

where $f_0$ is the center or nominal frequency, $\Delta f$ is the peak-to-peak value of the frequency deviation, and $f_m$ is the modulation frequency. Prior to the measurement by the ARPS, the Allan variance of frequency fluctuations of this controlled dithered test signal was calculated. The power spectral density of frequency fluctuations can be derived from (4), and expressed as

$$S_y(f) = \frac{\Delta f^2}{8 f_0^2} \delta(f - f_0) \tag{5}$$

where the suffix $y$ represents the frequency fluctuations normalized to $f_0$, and $\delta(f - f_0)$ is a delta function. The relation between the Allan variance and the power spectral density of frequency fluctuations is given by [10], [11]

$$\sigma_y^2(\tau) = 2 \int_0^\infty S_y(f) \frac{\sin^4(\pi f \tau)}{(\pi f \tau)^2} df. \tag{6}$$

The Allan variance for (4) can be calculated from (5) and (6) to give

$$\sigma_y^2(\tau) = \left( \frac{\Delta f \cdot \sin^2(\pi f_m \tau)}{2 \cdot f_0 \cdot \pi f_m \tau} \right)^2. \tag{7}$$

Fig. 5 shows the values of the square root of the Allan variance calculated by (7) as well as the values measured by the ARPS. The values of $f_m$, $\Delta f$, and $f_0$ in (4) and (7) were fixed to 12.3 kHz, 12 MHz, and 22 MHz, respectively. The good agreement of the two results in this figure confirms the accurate operation of the ARPS.

By using the ARPS, the frequency stability of two semiconductor lasers (CSP-type, Hitachi, HL8312E, 0.83

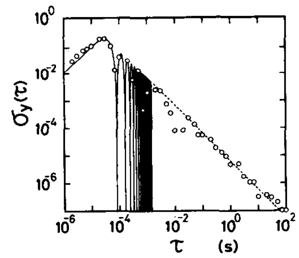


Fig. 5. Results of the operation test of the ARPS by using an artificially dithered signal. The solid curve and broken line represent calculated values, where the broken line is an envelope of the sinusoidally oscillating solid curve. Open circles represent measured values. The values of modulation frequency $f_m$, frequency deviation $\Delta f$, and nominal frequency $f_0$ were fixed to 12.3 kH, 12 MHz, and 22 MHz, respectively.

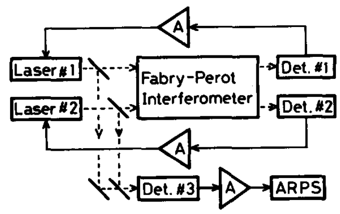


Fig. 6. Experimental setup for the frequency stability measurement of two semiconductor lasers whose frequencies were stabilized independently by using a Fabry–Perot interferometer as a common frequency reference.

$\mu$m wavelength) was measured. Fig. 6 shows the experimental setup. The frequencies of both lasers were independently stabilized by using a linear portion of a slope of the resonance curve of a Fabry–Perot interferometer as a common frequency reference. The Fabry–Perot interferometer was made using a cylindrical rod of fused silica with dielectric multilayer films coated on both ends. The reflectivity of these films was 90%. The free spectral range of this interferometer was 3.45 GHz. The Allan variance $\sigma_{yM}^2(\tau)$ of residual frequency fluctuations of the heterodyne signal between the two lasers was measured by the ARPS. The average of the heterodyne signal was 200 MHz, and this frequency was divided by 5 before it was applied to the ARPS. Fig. 7 shows the results of this measurement. In this figure, the square root of the Allan variance $\sigma_{yM}^2(\tau)$ of the residual frequency fluctuations $\delta f$ was represented by normalizing to the nominal optical frequency $f_0$, i.e., $y = \delta f / f_0$. Curves $A$ and $B$ represent the values under free-running and feedback conditions, respectively. Since the bandwidth of the feedback loop was 20 kHz, the value of curve $B$ was smaller than that of curve $A$ for the range of $\tau \geqq 50$ $\mu$s. The minimum of curve $B$ was $1.5 \times 10^{-10}$ at $\tau = 2$ s, which corresponds to frequency fluctuations of 54 kHz.

The frequency stability of the heterodyne signal in a HFLL was also measured. Fig. 8 shows an experimental setup for this measurement. Detailed explanations of this system has been given elsewhere [12]. The master laser frequency was stabilized by the method shown in Fig. 6, and its frequency stability was approximated by curve $B$ in Fig. 7. The frequency of the heterodyne signal between

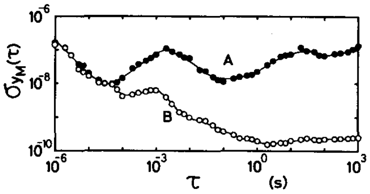


Fig. 7. Frequency stability of the semiconductor lasers measured by using the ARPS. $\sigma^2_{yM}(\tau)$ represents the square root of the Allan variance of the residual frequency fluctuations of the heterodyne signal between two semiconductor lasers, which was normalized to the nominal optical frequency. Curves $A$ and $B$ represent the values under free-running and feedback conditions, respectively.

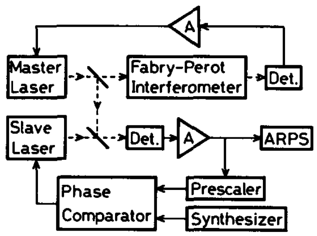


Fig. 8. Experimental setup of HFLL.

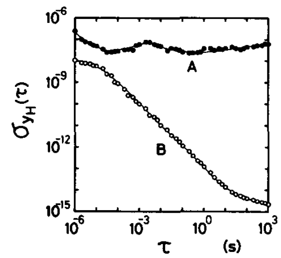


Fig. 9. Frequency stability of the heterodyne signal between the master and the slave lasers by using the ARPS. $\sigma^2_{yM}(\tau)$ represents the square root of the Allan variance of the residual frequency fluctuations of the heterodyne signal, which was normalized to the optical frequency. Curves $A$ and $B$ represent the values under free-running and feedback conditions, respectively.

the master and the slave lasers was locked to that of a stable microwave synthesizer so that the slave laser frequency accurately tracked the master laser frequency. The frequency of the heterodyne signal was 30 MHz. The Allan variance $\sigma^2_{yM}(\tau)$ of the residual frequency fluctuations of the heterodyne signal was measured by the ARPS to evaluate the performance of this system quantitatively. Fig. 9 shows the results of this measurement. This figure shows relative frequency fluctuations between master and locked slave lasers. These results are also expressed by the square root of the Allan variance $\sigma^2_{yM}(\tau)$ of the residual frequency fluctuations $\delta f$ normalized to the nominal optical frequency $f_0$ ($y = \delta f/f_0$). Curves $A$ and $B$ represent the values under free-running and feedback conditions, respectively. Since this system has a bandwidth wider than 1 MHz, the value of curve $B$ is lower than that of curve $A$ at $\tau \geq 1$ $\mu$s. The minimum of curve $B$ was 2.2 $\times 10^{-15}$ at $\tau = 1000$ s, which corresponds to relative frequency fluctuations as low as 0.8 Hz.

## V. Conclusion

An Allan variance ARPS was developed for frequency stability measurements of semiconductor lasers. The performance of this system is summarized as follows.

1) Since the number $C_i$ of the input TTL pulses in this system can be read without clearing the counter and there was no deadtime in measurements, and the Allan variance can be accurately measured.

2) The square root of the Allan variance can be measured for integration times from 1 $\mu$s to 10 000 s.

3) The maximum number of data of values of the averaged frequency $f_i$ measurable by this system is 707.

4) The highest measurable frequency of the signal is 90 MHz.

The result of measurements of the frequency stability of two semiconductor lasers by the ARPS gives a minimum $\sigma^2_{yM}(\tau)$ of $1.5 \times 10^{-10}$ at $\tau = 2$ s. This corresponds to frequency fluctuations of $\pm 54$ kHz. Measurements of the frequency stability of the heterodyne signal in the heterodyne frequency locked loop gave a minimum $\sigma^2_{yM}(\tau)$ of $2.2 \times 10^{-15}$ at $\tau = 1000$ s. This corresponds to frequency fluctuations between the locked laser and the master laser of about 0.8 Hz.


## Acknowledgment

The authors thank Dr. Y. Saburi, S. Kodato, S. Kinugawa, and T. Saito (Anritsu Co.), and T. Kato (Tokyo Institute of Technology) for their technical support.


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